# MUHAMMAD SOHAIL

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## PROFESSIONAL SUMMARY :

AI Engineer with a BS in Data Science and hands-on experience building and deploying machine learning and LLM-based applications. Skilled in Python, LangChain, and RAG pipeline development, with practical experience integrating vector databases (ChromaDB), LLM APIs (Groq, OpenAI), and REST APIs (FastAPI) into working applications. Comfortable across the full pipeline from data processing and model training to deployment with Streamlit and Docker.

## TECHNICAL SKILLS :

**AI/ML:** Supervised & Unsupervised Learning, Classification, Regression, Clustering, XGBoost, CatBoost, Random Forest, Model Evaluation
**LLM & Generative AI:** LangChain, RAG , OpenAI API, Groq API, LLM Integration, ChromaDB
**NLP:** Text Extraction & Processing, TF-IDF Vectorization, Text Classification
**Backend/API:** FastAPI, REST API Development, Pydantic, PostgreSQL Integration
**MLOps/Deployment:** Docker, Streamlit, GitHub Actions (CI/CD), Git/GitHub, Virtual Environments
**Programming:** Python, SQL
**Data/Databases:** Pandas, NumPy, Feature Engineering, EDA, Data Cleaning & Wrangling, PostgreSQL
**Visualization/Other:** Power BI, Matplotlib, Seaborn, Plotly, Statistics (Hypothesis Testing, A/B Testing, Distributions)

## AI/ML PROJECTS :

**Pakistan Tourist RAG System** | *LangChain, ChromaDB, Groq API, Streamlit*

- Built an end-to-end Retrieval-Augmented Generation (RAG) system that answers natural-language tourism queries about Pakistan using a custom knowledge base.
- Implemented the retrieval pipeline with LangChain, using ChromaDB as the vector store for semantic document search over tourism data.
- Integrated the Groq API as the LLM backend to generate context-grounded responses from retrieved documents, reducing hallucination compared to a standalone LLM.
- Deployed as an interactive Streamlit web app supporting real-time query answering, demonstrating a complete LLM application workflow from ingestion to user-facing deployment.

**Resume Scanner System** | *Python, NLP, Scikit-learn, Streamlit*

- Built an AI-powered resume screening application that extracts and parses text from PDF, DOCX, and TXT files.
- Applied NLP preprocessing and TF-IDF vectorization to convert unstructured resume text into model-ready features.
- Trained a classification model to predict job categories from resume content, automating a manual screening step.
- Deployed as an interactive Streamlit web application for real-time resume analysis.

**FastAPI ML API** | *FastAPI, Docker, GitHub Actions (CI/CD)*

- Exposed trained ML models as REST API endpoints using FastAPI, enabling programmatic integration with external applications.
- Containerized the service with Docker for consistent, portable deployment across environments.
- Set up a GitHub Actions CI/CD workflow to automate build and validation steps.

**Customer Churn Prediction App** | *Python, XGBoost, Streamlit, SMOTE*

- Built an end-to-end ML pipeline to predict customer churn, including class-imbalance handling with SMOTE.
- Trained and evaluated an XGBoost classification model, then deployed it via Streamlit for interactive, business-facing churn risk analysis.

**Liver and Heart Disease Prediction** | *Python, Scikit-learn, Streamlit*

- Developed an end-to-end classification project covering data collection, cleaning, EDA, and feature engineering for two separate disease-prediction tasks.
- Trained and evaluated multiple classification models, visualized key insights, and deployed the final models as an interactive Streamlit web app.

**Weather Forecast Analysis Dashboard** | *Power BI, SQL*

- Built a Power BI dashboard using SQL-based data extraction, with KPIs and interactive visualizations for weather trend analysis.

## EDUCATION :

**Government College University Faisalabad (GCUF)** 2021 – 2025

- Bachelor's in Data Science